

## UI/UX DESIGN DOCUMENT

# Musanze Safe Market

## Field Inspection Prototype

MOB\_A1\_G15\_UIUX.pdf

|                         |                                           |
|-------------------------|-------------------------------------------|
| Group Verification Code | MOB-G15-7861                              |
| Group Number            | G15                                       |
| Submission Date         | 30 September 2026                         |
| Course                  | SWE 3409 — Mobile Application Development |
| Assignment              | Assignment 1                              |

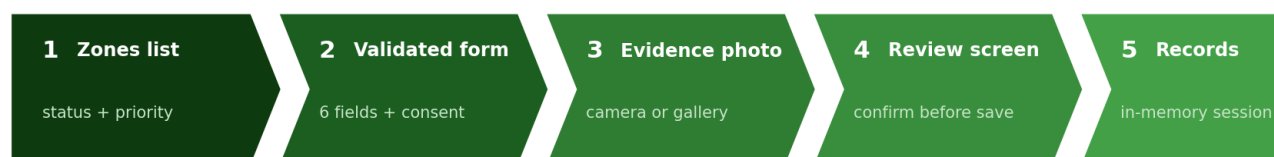
| # | Group member          | Reg. number | Role                              |
|---|-----------------------|-------------|-----------------------------------|
| 1 | Munyaneza Ngabo       | 25/27861    | Navigation Architect & UI/UX Lead |
| 2 | OBAME Videlina ketsia | 25/27671    | Camera & Media                    |
| 4 | Halime Brahim Maye    | 25/28258    | State & Validation                |
| 5 | Umubyiye Marie Merci  | 26/38496    | Release Evidence                  |

### Problem Statement

Musanze Safe Markets is running a one-day pilot to register vendors and document visible stall conditions in assigned market zones. Field officers currently rely on paper forms, which are slow, error-prone, and lose evidence photographs. The Musanze Safe Market prototype replaces the paper workflow with a mobile application that:

1. Lists assigned market zones with status and inspection priority.
2. Captures a fictional vendor inspection with validated input.
3. Attaches one evidence photograph (camera or gallery).
4. Displays a review screen for confirmation before saving.
5. Stores records in memory for the duration of the session.

### Solution at a glance



**The app works offline, requires no backend, and never collects real personal data.**

## User and Task Flow

### 2.1 Primary User

**Jean, 28**

Field Officer

Works outdoors, walks between stall zones, holds the phone in one hand while inspecting stalls. Needs large touch targets, minimal typing, and clear status feedback in sunlight.

| Goals                                                                                                                                                                                                                                                                        | Constraints                                                                                                                                                                                                                                       |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <ul style="list-style-type: none"> <li>Complete an inspection in under 2 minutes.</li> <li>Capture a photo without leaving the form.</li> <li>See immediately which fields failed validation.</li> <li>Confirm the saved record contains the correct information.</li> </ul> | <ul style="list-style-type: none"> <li>Small phone screen (360 × 640 logical pixels or larger).</li> <li>One-handed use while walking.</li> <li>Glare and outdoor lighting.</li> <li>Spotty network — the app must work fully offline.</li> </ul> |

### 2.2 Complete Screen Flow

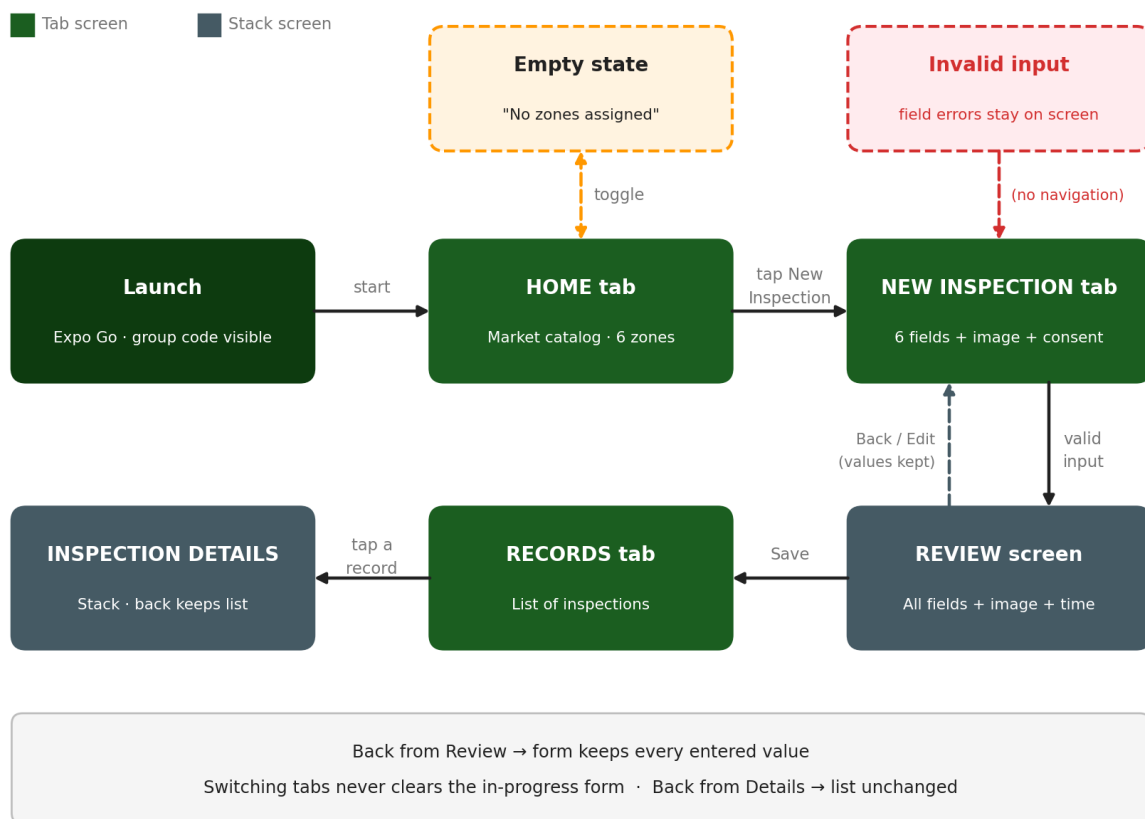
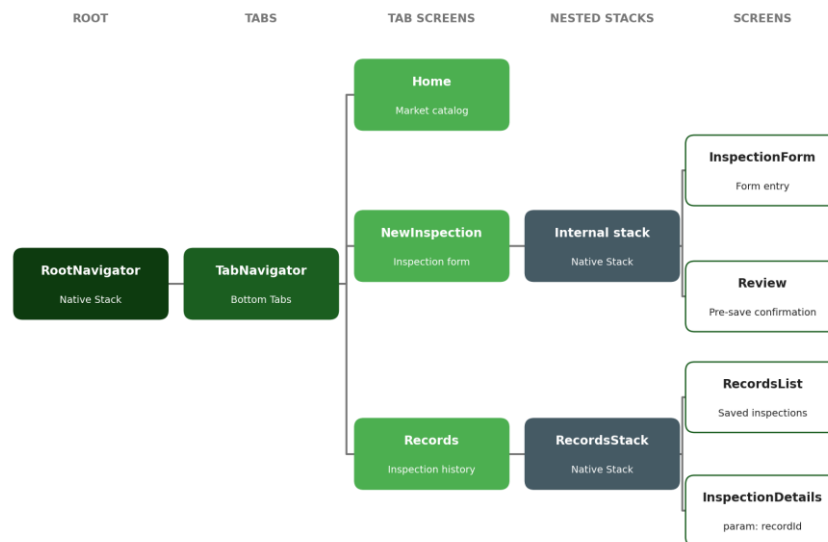


Figure 1 — Screen flow from launch to saved record. Dashed arrows are alternative or return paths.

## Information Architecture

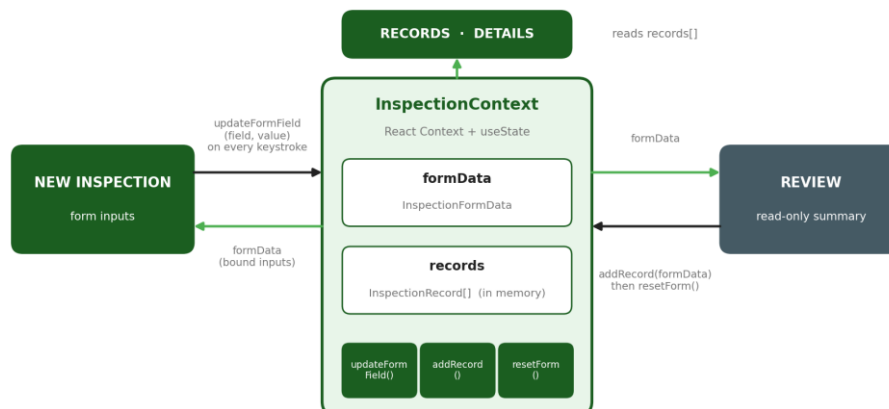
### 3.1 Navigation Tree



### 3.2 Route Names and Data Passing

| Route name        | Params               | Purpose                           |
|-------------------|----------------------|-----------------------------------|
| Home              | none                 | Market zone list                  |
| NewInspection     | none                 | Form entry                        |
| Review            | none (reads context) | Review before saving              |
| Records           | none                 | List all records                  |
| RecordsList       | none                 | Record list root                  |
| InspectionDetails | { recordId: string } | Detail view of a saved inspection |

### 3.3 Data Flow



Form state lives at context level so that back navigation from Review preserves input, switching tabs does not reset the in-progress inspection, and the Records list updates immediately after saving. No persistence, no network — the assignment excludes both.

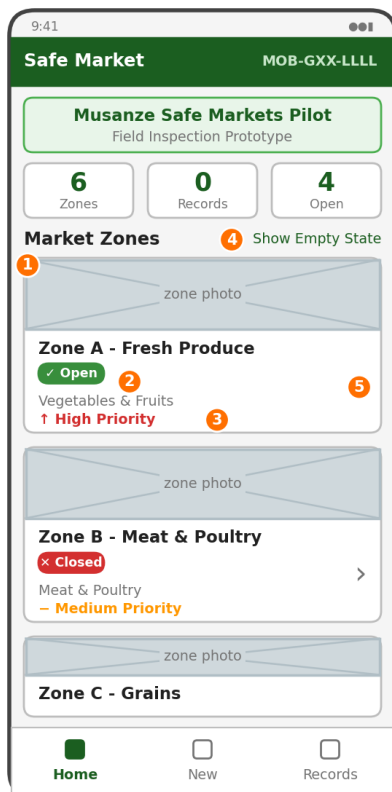
### 3.4 Back Behavior

- Records → InspectionDetails → Back: returns to the Records list, unchanged (state preserved in context).
- NewInspection → Review → Back: returns to the form with all previously entered values still present.
- Tab switches never clear the in-progress form.

## Wireframes (Annotated Low-Fidelity)

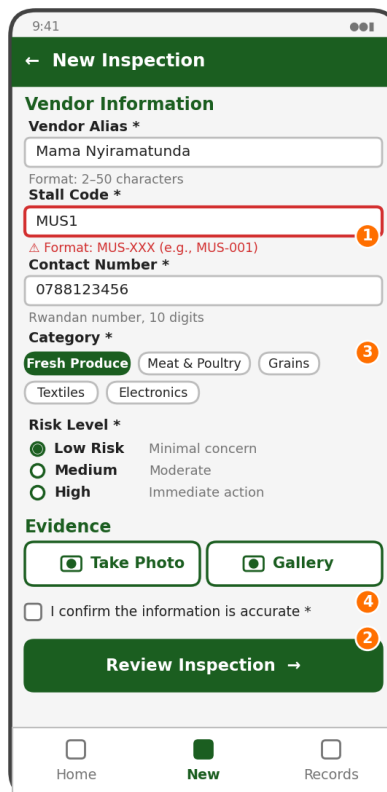
Orange numbered markers on each wireframe correspond to the notes beneath it.

### 4.1 Home Screen (Catalog)



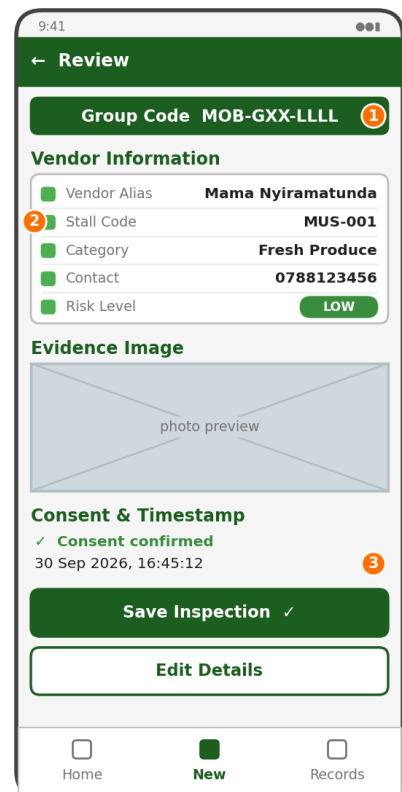
- 1 Cards use the reusable MarketCard component.
- 2 Status chip is colour-coded (green / red / amber) and carries an icon.
- 3 Priority badge uses arrow icon + text, never colour alone.
- 4 Toggle link reveals the empty state for the demo.
- 5 Minimum touch target: 48px on the row and chips.

### 4.2 New Inspection Form



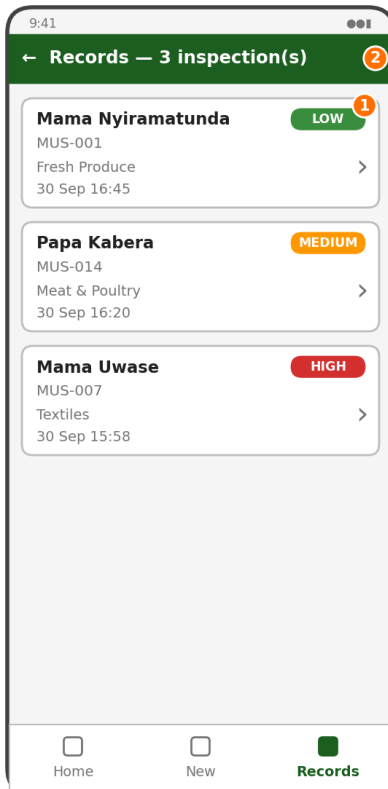
- 1 Errors appear immediately below the offending field.
- 2 Blocked submission: the primary button does not navigate while any field error is present.
- 3 Pill selector avoids a native dropdown that is hard to use one-handed.
- 4 Consent checkbox is required for legal/ethical reasons.

### 4.3 Review Screen



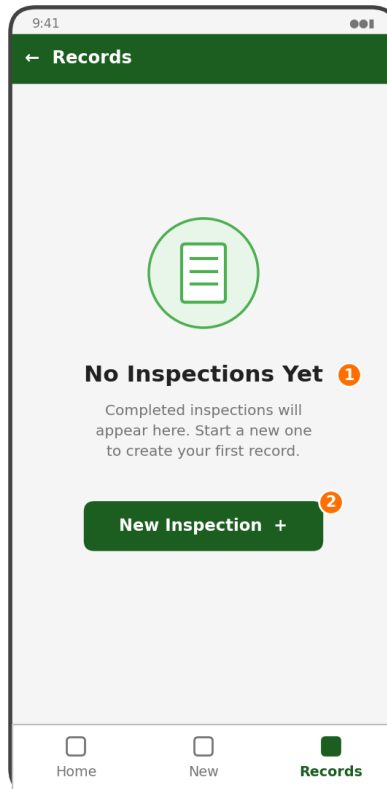
- 1 Group verification code sits in a coloured banner at the top.
- 2 Every field is read-only and shown with an icon for scanability.
- 3 Timestamp uses 24-hour format (Africa/Kigali).

## 4.4 Records List



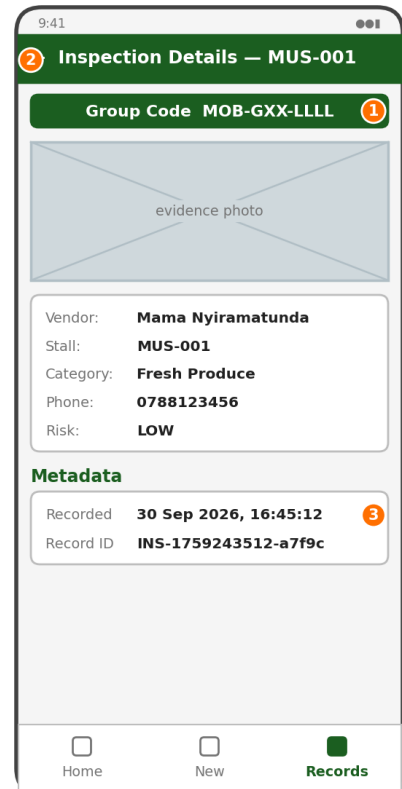
- 1 Tap a card to open Inspection Details.
- 2 Header shows the live count of saved inspections.

## 4.5 Empty State (no records)



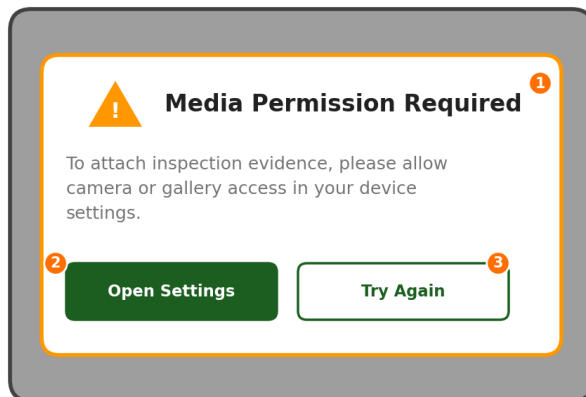
- 1 Shown when no records exist; explains what will appear here.
- 2 One clear next action: start a new inspection.

## 4.6 Inspection Details (Stack)



- 1 Group verification code banner repeated for traceability.
- 2 Back returns to the Records list, unchanged.
- 3 Metadata: timestamp and generated record ID.

## 4.7 Camera Permission Denied (Recovery State)



## Recovery path

- 1 Plain-language explanation of why access is needed.
- 2 Open Settings takes the officer straight to the device permission page.
- 3 Try Again re-requests access without leaving the form.

## Visual Design

### 6.1 Color Palette

| Swatch  | Token             | Usage                            |
|---------|-------------------|----------------------------------|
| #1B5E20 | primary           | Headers, primary buttons, focus  |
| #4CAF50 | primaryLight      | Success icons, empty-state icon  |
| #0D3B0F | primaryDark       | Pressed state on primary buttons |
| #FF6F00 | secondary         | Secondary actions                |
| #F5F5F5 | background        | Screen background                |
| FFFFFF  | surface           | Cards, inputs                    |
| #212121 | text              | Primary body text                |
| #757575 | textSecondary     | Hints, captions                  |
| #D32F2F | error             | Validation errors                |
| #FF9800 | warning           | Medium risk, permission notice   |
| #388E3C | success           | Consent confirmed, status open   |
| #388E3C | statusOpen        | Green status chip                |
| #D32F2F | statusClosed      | Red status chip                  |
| #FF9800 | statusMaintenance | Amber status chip                |

Contrast check (WCAG 2.1) — ratios computed from the hex values

| Sample         | Pair                          | Ratio  | Level                |
|----------------|-------------------------------|--------|----------------------|
| Aa Sample text | primary on surface            | 7.9:1  | AAA                  |
| Aa Sample text | error on errorLight (#FFEBEE) | 4.4:1  | AA (large text only) |
| Aa Sample text | text on surface               | 16.1:1 | AAA                  |

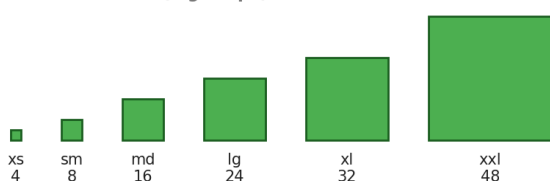
### 6.2 Typography Scale

| Style   | Sample (scaled) | Size / Weight / Line | Usage                |
|---------|-----------------|----------------------|----------------------|
| h1      | Safe Market     | 28 / 700 / 36        | Screen hero (unused) |
| h2      | Safe Market     | 24 / 700 / 32        | Header title         |
| h3      | Safe Market     | 20 / 600 / 28        | Card title, section  |
| body    | Safe Market     | 16 / 400 / 24        | Paragraph, inputs    |
| bodySm  | Safe Market     | 14 / 400 / 20        | Card secondary text  |
| caption | Safe Market     | 12 / 400 / 16        | Hints, meta          |
| label   | Safe Market     | 14 / 600 / 20        | Field labels         |
| button  | Safe Market     | 16 / 600 / 24        | Buttons              |

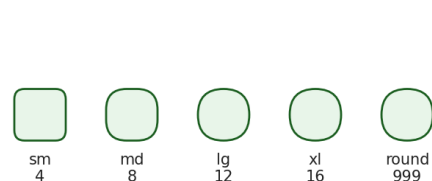
System font stack only — no custom fonts, to keep the Expo Go build identical on iOS and Android.

### 6.3 Spacing, Radii and Elevation

SPACING SCALE (logical px)



BORDER RADII



Elevation: sm — subtle (cards) · md — raised buttons · lg — modals (unused in this iteration).

## 6.4 Reusable Component Inventory

| Component         | Location              | Reused in      |
|-------------------|-----------------------|----------------|
| ScreenContainer   | components/common     | All screens    |
| AppHeader         | components/common     | All screens    |
| PrimaryButton     | components/common     | Form, Review   |
| EmptyState        | components/common     | Home, Records  |
| FormField         | components/common     | New Inspection |
| MarketCard        | components/market     | Home           |
| StatusChip        | components/market     | MarketCard     |
| PriorityBadge     | components/market     | MarketCard     |
| CategorySelector  | components/inspection | New Inspection |
| RiskLevelSelector | components/inspection | New Inspection |
| ConsentCheckbox   | components/inspection | New Inspection |
| ImagePicker       | components/media      | New Inspection |
| ImagePreview      | components/media      | ImagePicker    |
| PermissionNotice  | components/media      | ImagePicker    |

## 6.5 Final Screen Mockups

Paste the six screenshots from the running Expo Go app into the boxes below (right-click → Change Picture, or delete the label and insert). They are identical to the submitted build.

|                                                            |                                                        |                                                            |
|------------------------------------------------------------|--------------------------------------------------------|------------------------------------------------------------|
| <b>01-home.png</b><br><i>Insert screenshot here</i>        | <b>02-catalog.png</b><br><i>Insert screenshot here</i> | <b>03-empty-state.png</b><br><i>Insert screenshot here</i> |
| <b>04-form-errors.png</b><br><i>Insert screenshot here</i> | <b>05-review.png</b><br><i>Insert screenshot here</i>  | <b>06-details.png</b><br><i>Insert screenshot here</i>     |

## Accessibility and Design Decisions

### 8.1 Accessibility

|          |                                                                |                      |
|----------|----------------------------------------------------------------|----------------------|
| STATUS   | <div>✓ Open</div> <div>✕ Closed</div> <div>⚠ Maintenance</div> | icon + text + colour |
| PRIORITY | <div>↑ High</div> <div>– Medium</div> <div>↓ Low</div>         | never colour alone   |

|               |                                                                                                                                                                                                                                                            |
|---------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Touch targets | <ul style="list-style-type: none"> <li>All interactive elements are at least 44 × 44 logical pixels.</li> <li>Primary buttons use paddingVertical: 14 plus a 16px font.</li> </ul>                                                                         |
| Contrast      | <ul style="list-style-type: none"> <li>Body text (#212121) on white: 16.1:1 (AAA).</li> <li>Primary (#1B5E20) on white: 7.9:1 (AAA).</li> <li>Chip text on light backgrounds: minimum 4.5:1 across all three statuses and all three priorities.</li> </ul> |



|                                  |                                                                                                                                                                                                                                                                                                |
|----------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Colour independence</b>       | <ul style="list-style-type: none"> <li>• Status is never colour alone: each chip has a distinct icon (check / close / construct).</li> <li>• Priority is never colour alone: each badge has an arrow (up / remove / down).</li> </ul>                                                          |
| <b>Error messaging</b>           | <ul style="list-style-type: none"> <li>• Every field has a field-level error message directly below it.</li> <li>• Errors are instructions, not accusations: “Format: MUS-XXX (e.g., MUS-001)”, not “Invalid input”.</li> <li>• Submission is blocked until all errors are cleared.</li> </ul> |
| <b>Screen readers</b>            | <ul style="list-style-type: none"> <li>• Every input uses a visible label, never a placeholder-only field.</li> <li>• Icon buttons include a text label (“Take Photo”, “Choose from Gallery”).</li> </ul>                                                                                      |
| <b>Large font / narrow width</b> | <ul style="list-style-type: none"> <li>• Flex layouts, no fixed widths; text uses system font scaling and wraps at word boundaries.</li> <li>• Verified at 320px width and 1.3× font scale.</li> </ul>                                                                                         |

## 8.2 Three Justified Design Decisions

| Decision                                    | Why                                                                                                                                                                                                                 | Trade-off                                                                      |
|---------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------|
| <b>1 Bottom tabs, not a drawer</b>          | The primary user works one-handed, walking between stalls. A bottom tab bar puts all three destinations (Home, New Inspection, Records) inside thumb reach. A drawer needs a top-corner reach and an extra gesture. | Only three destinations fit comfortably — acceptable for the pilot scope.      |
| <b>2 Pills instead of a native picker</b>   | A native dropdown hides the category list behind an extra tap, and the Android wheel picker is awkward one-handed. Eight category pills are visible at once, so the officer selects without leaving the form.       | Eight pills use more vertical space — offset by a wrapping container.          |
| <b>3 Review screen owns the save action</b> | Saving directly from the form risks data-entry errors that are hard to undo (no backend, no persistence). Review shows every value read-only and requires an explicit Save tap.                                     | Adds one screen and one tap — the officer sees exactly what is being recorded. |

## 8.3 Known Limitations

- Records are stored in memory only and are lost when the app closes — intentional, matching the assignment scope (no backend, no persistent storage).
- The app uses fictional vendor aliases only; no real personal data is collected, transmitted, or stored.
- Images are stored as local URIs and are not uploaded anywhere.

## 8.4 File Mapping (UI/UX PDF ↔ Implemented App)

| UI/UX section             | Implemented in                                  |
|---------------------------|-------------------------------------------------|
| <b>Home / Catalog</b>     | src/screens/Home/HomeScreen.tsx                 |
| <b>Inspection form</b>    | src/screens/Inspection/NewInspectionScreen.tsx  |
| <b>Review</b>             | src/screens/Inspection/ReviewScreen.tsx         |
| <b>Records list</b>       | src/screens/Records/RecordsScreen.tsx           |
| <b>Inspection details</b> | src/screens/Records/InspectionDetailsScreen.tsx |
| <b>Media workflow</b>     | src/components/media/ImagePicker.tsx            |
| <b>Theme tokens</b>       | src/theme/colors.ts, typography.ts, spacing.ts  |
| <b>Group code display</b> | src/components/common/AppHeader.tsx             |